

More continuous random variables

1. Imagine picking a real number between 0 and 10 so that (roughly speaking) every number is equally likely. Your number is a random variable X , which we describe using the density

function f defined by $f(x)=0$, if $x<0$; $f(x)=\frac{1}{10}$, if $0\leq x\leq 10$; $f(x)=0$, if $x>10$.

- a. Sketch a graph of this density function.

- b. Calculate $P(3\leq X\leq 7)$. Shade the part of your graph corresponding to the event $(3\leq X\leq 7)$.

- c. The expected value of X can be calculated as $\int_0^{10} xf(x)dx$. This integral looks like the sum $(\sum_x xf(x))$, where x runs over the values of X that we use to calculate expected value of a discrete random variable. For a continuous random variable, we replace $f(x)$ with $f(x)dx$ and the sum with an integral. Calculate $E[X]$.

- d. The Law of the Unconscious Statistician also works for continuous random variables.

Calculate $E[X^2]$ (second moment of X) as $\int_0^{10} x^2 f(x)dx$. Then calculate $\text{Var}(X)$ as $E[X^2]-E[X]^2$.

2. Imagine you pick two numbers independently at random from the uniform distribution on $[0, 10]$. That is the distribution from the previous question where, roughly speaking, every number in $[0, 10]$ is an equally likely pick. Let Y be the higher of the two numbers that you pick. The density function f for Y has the following formula: $f(x)=0$, if $x<0$; $f(x)=x/50$, if $0\leq x\leq 10$; $f(x)=0$, if $x>10$.
- a. Sketch a graph of this density function.
- b. Use this density function to calculate $P(8\leq Y\leq 10)$, the probability that the higher of the two numbers is at least 8. Shade the part of your graph corresponding to the event $(8\leq Y\leq 10)$.
- c. Calculate $E[Y]$.
- d. Calculate $E[Y^2]$ (the second moment) and $\text{Var}(Y)$.